

Ayush Mahale

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Education

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science and Statistics; **GPA: 3.85/4.0**

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms (CS61B), AI & ML (CS 189), Computer Architecture (CS 61C), Computer Graphics (CS 184), Graduate Network Theory (CS 294), Object Oriented Programming (CS 61A), Principles and Techniques in Data Science (DATA C100), Graduate Bayesian Statistics (STAT 238), Discrete Math and Probability Theory (CS 70), Random Processes (EECS 126), Theoretical Statistics (STAT 135), Linear Algebra & Differential Eqs (MATH 54), Calculus (MATH 51/52/53)

Experience

SciFin Inc.

San Jose, CA

Software Engineering Intern

May 2025 – Aug 2025

- Led the intern team of 4 in building a Go-based system (Genesis) currently used by all teams to generate large-scale synthetic Salesforce CRM data (Accounts, Contacts, Opportunities, etc.) that reduced pipeline running time by 30% down to 8 minutes.
- Built Meltano ELT pipelines to ingress generated data into MongoDB and Salesforce, handling schema mappings and validations.
- Set up k8s framework and integrated the project into the company's k8s cluster, allowing project to be run alongside the existing architecture.
- Gave a technical demo of Genesis project to 4 VCs at Foundation Capital.

Computer Science Mentors

Berkeley, CA

CS70 Mentor

Sep 2024 – Dec 2024

- Led tutoring sections on Discrete Math and Probability Theory for 7 students twice a week, earning perfect overall rating from students (top 10%).
- Worked with a group of fellow mentors to organize monthly social events across CS courses for 50+ students.

Haas School of Business, UC Berkeley

Berkeley, CA

Teaching Assistant

Sep 2024 – Feb 2025

- Taught and led discussions about Finance & Entrepreneurship for 60+ students twice a week.
- Coordinated and assisted with team building and leadership activities.
- Worked alongside Program Manager to design the curriculum for 100+ incoming Fall 2025 batch of students.

Projects

Cloth Simulation Engine | ([Website](#))

April 2026

- Built a physically-based cloth simulator using a mass-spring model with Verlet integration, enabling realistic deformation such as sagging, folding, and collisions.
- Implemented collision handling (spheres, planes, self-collisions) and GPU shading techniques to render high-quality, dynamic cloth behavior.

RISC-V Classifier ML Pipeline

March 2026

- Built an end-to-end handwritten digit classifier in RISC-V assembly, implementing core linear algebra ops and neural network components.
- Managed low-level memory, file I/O, and function calls under strict calling conventions, ensuring correctness across multi-function pipelines.

Paulownia Advisory, CTO

Sep 2023 – Oct 2024

- Led the use of financial modeling tools to streamline firm's investment process (\$60k+ assets under management).
- Developed and built the firm's digital platform using WIX Sitebuilder.

Build Your Own World | ([GitHub](#))

Oct 2024

- Built a mobile/web application in Java where user can control a character and interact in a randomly generated world, with integrated features like minigames.

ZvezdaCardio, CalHacks | ([GitHub](#))

Jul 2024

- Developed a mobile application in Swift & JS which tracks user steps & calories burnt with a generative AI chatbot trained on live data (Gemini) offering personalized insights to users.

Technical Knowledge

Programming Languages: Java, Python, Go, C, JavaScript, Swift, Scheme, SQL

Frameworks & Tools: Git, Jupyter, Eclipse, IntelliJ, VSCode, Cursor, Claude Code